

MSDS 535 – Week 15

A.1, A.2

Overview and Objectives

- Probability and statistics.
- Numerical optimization.

PDF of Typical Distributions

- Probability density functions (PDFs) describe the likelihood of a continuous random variable equating a certain sample.
- PDFs must be non-negative for all samples and integrate to 1 over the entire sample space.
- Typical distributions like Uniform, Exponential, Normal, Gamma, and Beta have specific PDFs.
- Each distribution has defined mean, variance, and formulas for their respective PDFs.

PDF of Typical Distributions

- Understanding these distributions aids in modeling real-world phenomena and statistical analysis.
- The Exponential Distribution, for instance, models the time between events in a Poisson process.
- The Beta Distribution finds applications in Bayesian inference and modeling proportions.

MLE and MAP

- Maximum Likelihood Estimation (MLE) and Maximum A Posteriori (MAP) estimate parameters from observed data.
- MLE assumes fixed parameters and maximizes the likelihood of observing the data given those parameters.
- MAP treats parameters as random variables, considering prior probability distributions.

MLE and MAP

- These methods play a crucial role in various fields like machine learning, statistics, and signal processing.
- MLE and MAP offer distinct perspectives on parameter estimation based on different assumptions.
- Understanding their differences helps in choosing appropriate estimation techniques for different scenarios.
- Both methods are foundational in statistical inference and model fitting.

Significance Test

- A procedure to assess claims about observed data using null and alternative hypotheses.
- Significance level (α) determines the probability of mistakenly rejecting the null hypothesis when it's true.
- The p-value indicates the strength of evidence against the null hypothesis.
- Significance testing is vital in scientific research, enabling researchers to draw valid conclusions.

Significance Test

- Different significance levels (e.g., 0.001, 0.005, 0.01, 0.05) offer varying degrees of confidence.
- Statistical methods like z-test and t-test provide tools for hypothesis testing.
- Understanding significance tests is crucial in decision-making processes and drawing accurate conclusions.

Density Estimation

- Density Estimation estimates a probability distribution from observed data.
- Nonparametric methods (Histogram, Naive Estimator, Kernel Estimator, Nearest Neighbor) and Parametric methods (based on known distributions) are used.
- Each method has distinct approaches to estimate probability density functions.
- Density estimation is fundamental in various fields like pattern recognition and machine learning.

Density Estimation

- Nonparametric methods allow flexibility without assumptions about underlying distributions.
- Parametric methods assume a specific distribution form and estimate its parameters.
- Choosing the right method depends on the nature of the data and the assumptions made.

Bias-Variance Tradeoff

- Bias represents the difference between an estimator's expected prediction and the true values.
- Variance measures the variability of an estimator's predictions.
- Balancing bias and variance is crucial; increasing model complexity may decrease bias but increase variance, leading to overfitting.
- The tradeoff impacts model performance and generalization to unseen data.

Bias-Variance Tradeoff

- Various techniques like regularization aim to strike a balance between bias and variance.
- Understanding the bias-variance tradeoff aids in model selection and performance optimization.
- It's a critical concept in machine learning and helps in designing robust predictive models.

Cross-validation and Jackknife

- Cross-validation divides data into subsets for model validation and evaluation, crucial for scarce data.
- Jackknife constructs confidence intervals by computing leave-one-out residuals to avoid overfitting issues.
- Both techniques contribute to improving model reliability and generalization.

Cross-validation and Jackknife

- Cross-validation helps in selecting models with better predictive performance.
- Jackknife's approach reduces bias in estimating model performance.
- Proper utilization of these techniques enhances the reliability of statistical analyses.
- Understanding their mechanics is pivotal in ensuring accurate model assessment and validation.

Batch Gradient Descent

- Minimizing a loss function with the gradient of model parameters guides updates.
- Batch gradient descent computes gradients on the entire dataset for updates.
- It's an effective yet computationally intensive method for optimization.
- Updates are determined by the learning rate and the direction of the gradient.
- Iteratively adjusts parameters to find the local minima of the loss function.

Stochastic Gradient Descent

- Evaluates the loss function on individual data points for updates.
- Reduces computational cost compared to batch gradient descent.
- Adds randomness to avoid local minima in optimization.
- Balances efficiency and effectiveness in parameter updates.
- Mini-batch gradient descent combines advantages of both approaches.

Momentum

- Prevents oscillation in updates by considering previous update vectors.
- Works in batch, mini-batch, or stochastic styles based on data point selection.
- Uses a fraction (γ) of the previous update and the current gradient.
- Helps overcome local minima and accelerates convergence.

Adagrad

- Adjusts the learning rate adaptively based on past gradients.
- Utilizes a diagonal matrix to change the learning rates for each parameter.
- Prevents large gradients from dominating the learning process.
- Incorporates a hyperparameter to prevent divide-by-zero errors.

Adadelata

- Modifies Adagrad by decaying the sum of squared gradients exponentially.
- Uses root mean square of gradients to update parameters.
- Adapts learning rates for each parameter based on historical gradients.
- Helps in cases where Adagrad's aggressive diminishing of learning rates is problematic.

RMSprop

- Similar to Adadelta, dynamically adjusts learning rates based on gradients.
- Updates parameters using a moving average of squared gradients.
- Developed independently from Adadelta but shares similar principles.
- Mitigates the Adagrad issue of aggressive learning rate reduction.

Adam

- Estimates gradients' first and second moments for adaptive learning rates.
- Incorporates momentum and RMSprop features into its update mechanism.
- Balances between adaptive learning rates and momentum for efficient convergence.

AdaMax

- A variant of Adam focusing on infinity norm of moment vectors.
- Computes moment vectors using exponential moving averages.
- Utilizes infinity norm for parameter updates in optimization.

Newton's Method

- Aims to find the root of a function or the minimum of a loss function.
- Updates parameters by considering the inverse of the Hessian matrix.
- Works well in convex problems but computationally expensive.

Quasi-Newton Method

- Approximates the Hessian matrix for optimization without its direct computation.
- Utilizes a sequence of rank-one matrices to approximate the Hessian inverse.
- Balances computational cost and accuracy in solving optimization problems.

Coordinate Descent

- Solves optimization problems by minimizing one parameter at a time.
- Iteratively minimizes the loss function with respect to each parameter.
- Efficient for problems like sparse coding where the objective function is not convex.

Quadratic Programming

- Deals with quadratic objective functions and affine constraints.
- Presents solutions for equality-constrained quadratic programs (EQPs).
- Focuses on minimizing quadratic objectives subject to linear equality constraints.
- Offers versatile solutions based on various constraints and objective functions.